207 EDA

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2026-03-01

```r
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.2
```

```r
library(haven)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
com_students <- read_sav("Comparison_Students.sav")
```

```
## Failed to find G1READ_A
## Failed to find G1MATH_A
## Failed to find G2READ_A
## Failed to find G3READ_A
## Failed to find G3MATH_A
## Failed to find FILTER_$
```

```r
com_students
```

```
## # A tibble: 1,780 × 56
##    stdntid gender     race       preschol kindrgrt birthmonth birthday birthyear
##      <dbl> <dbl+lbl>  <dbl+lbl>  <dbl+lb> <dbl+lb>      <dbl>    <dbl>     <dbl>
##  1   30001 0 [MALE]   NA          0 [NO]   1 [YES]         11       22      1979
##  2   30002 0 [MALE]   NA          0 [NO]   1 [YES]          8        5      1980
##  3   30003 0 [MALE]     1 [WHITE] NA       1 [YES]         NA       NA        NA
##  4   30004 0 [MALE]     1 [WHITE]  0 [NO]  1 [YES]         NA       NA        NA
##  5   30005 0 [MALE]   NA          0 [NO]   1 [YES]          3       17      1980
##  6   30006 0 [MALE]   NA          0 [NO]   1 [YES]          7       16      1980
##  7   30007 0 [MALE]   NA         NA       NA               NA       NA        NA
##  8   30008 0 [MALE]   NA          0 [NO]   1 [YES]          1        2      1980
##  9   30009 1 [FEMALE] NA          0 [NO]   1 [YES]          3       30      1979
## 10   30010 0 [MALE]     1 [WHITE] NA       1 [YES]          1       19      1980
## # ℹ 1,770 more rows
## # ℹ 48 more variables: flagg1c <dbl+lbl>, flagg2c <dbl+lbl>, flagg3c <dbl+lbl>,
## #   g1schid <dbl>, g1tchid <dbl>, g1classsize <dbl>, g1treadss <dbl>,
## #   g1tmathss <dbl>, g1tlistss <dbl>, g1wordskillss <dbl>, g1readbsraw <dbl>,
## #   g1mathbsraw <dbl>, g1readbspct <dbl>, g1mathbspct <dbl>,
## #   g1readbsobjpct <dbl>, g1mathbsobjpct <dbl>, g2schid <dbl>, g2tchid <dbl>,
## #   g2classsize <dbl>, g2treadss <dbl>, g2tmathss <dbl>, g2tlistss <dbl>, …
```

```
high_schools <- read_sav("STAR_High_Schools.sav")
```

```
## Failed to find CITYPOP
## Failed to find COUNTYPO
## Failed to find SCHOOLTY
## Failed to find HIGHGRAD
## Failed to find GRADELEV
## Failed to find MINORITY
## Failed to find MINORI_A
## Failed to find MINORI_B
## Failed to find ENROLLME
## Failed to find MEALNUM
## Failed to find MEANPCT
## Failed to find MEALESTI
## Failed to find MINREQUI
## Failed to find MATHCRGR
## Failed to find SCIENCCR
## Failed to find FORLANGC
## Failed to find SOCSTUDC
## Failed to find COMPUTER
## Failed to find ENGLISHC
## Failed to find MATHCRUN
## Failed to find SCIENCEC
## Failed to find FORLAN_A
## Failed to find SOCSTU_A
## Failed to find COMPUT_A
## Failed to find ENGLIS_A
## Failed to find NOTGRADN
## Failed to find UNIFIEDP
## Failed to find LINEARAL
## Failed to find OTHERMAT
## Failed to find COURSE1
## Failed to find COURSE2
## Failed to find COURSE3
## Failed to find COURSE4
## Failed to find COURSE5
## Failed to find COURSE6
## Failed to find COURSE7
## Failed to find COURSE8
## Failed to find COURSE9
## Failed to find COURSE10
## Failed to find COURSE11
## Failed to find COURSE12
## Failed to find COURSE13
## Failed to find COURSE14
## Failed to find COURSE15
## Failed to find COURSE16
## Failed to find LANG01
## Failed to find LLEVEL01
## Failed to find HLEVEL01
## Failed to find LANG02
## Failed to find LLEVEL02
## Failed to find HLEVEL02
## Failed to find LANG03
```

```
## Failed to find LLEVEL03
## Failed to find HLEVEL03
## Failed to find LANG04
## Failed to find LLEVEL04
## Failed to find HLEVEL04
## Failed to find LANG05
## Failed to find LLEVEL05
## Failed to find HLEVEL05
## Failed to find LANG06
## Failed to find LLEVEL06
## Failed to find HLEVEL06
```

```
high_schools
```

```
## # A tibble: 161 × 33
##      hsid SCHLURBN     ENRLMENT seniors lowgradelevel HGHGRADE NUMGRADE MNRTYPCT
##     <dbl> <dbl+lbl>       <dbl>   <dbl>         <dbl>    <dbl>    <dbl>    <dbl>
##  1 106017 3 [SUBURBAN]     1500   375               9       12        4     2
##  2 112032 4 [RURAL]         400    66.7             7       12        6     5
##  3 112034 2 [URBAN]        1100   275               9       12        4    35
##  4 112036 4 [RURAL]         428   107               9       12        4     4.1
##  5 118044 4 [RURAL]         450    37.5             1       12       12     0
##  6 123053 4 [RURAL]         468   117               9       12        4     4.91
##  7 128073 3 [SUBURBAN]     1505   376.              9       12        4    10
##  8 128081 3 [SUBURBAN]     1233   308.              9       12        4     1.05
##  9 129084 3 [SUBURBAN]      435   109.              9       12        4    32
## 10 130086 3 [SUBURBAN]     1240   310               9       12        4     3
## # ℹ 151 more rows
## # ℹ 25 more variables: FRLCHPCT <dbl>, NOGRDPCT <dbl>, MINRQMNT <dbl+lbl>,
## #   MINMATH <dbl>, MINSCIEN <dbl>, MINFORLG <dbl>, MINSOCST <dbl>,
## #   MINCOMP <dbl>, MINENGLS <dbl>, algebra3 <dbl+lbl>, math4 <dbl+lbl>,
## #   precalculus <dbl+lbl>, calculus <dbl+lbl>, probability <dbl+lbl>,
## #   trigonometry <dbl+lbl>, analytical <dbl+lbl>, solidgeo <dbl+lbl>,
## #   LINALGBR <dbl+lbl>, FRENCH <dbl>, FREHILVL <dbl>, SPANISH <dbl>, …
```

```
k3_schools <- read_sav("STAR_K-3_Schools.sav")
```

```
## Failed to find SCHOOLTY
## Failed to find GRADERAN
## Failed to find ENROLLME
## Failed to find ATTENDAN
## Failed to find MEMBERSH
## Failed to find CHAPTER1
## Failed to find LUNCH
## Failed to find BUSED
## Failed to find NATIVEAM
## Failed to find ASIAN
## Failed to find BLACK
## Failed to find HISPANIC
## Failed to find WHITE
## Failed to find OTHERRAC
## Failed to find GRADEFIL
```

```
k3_schools
```

```
## # A tibble: 80 × 53
##     schid SCHLURBN    GRDRANGE FLAGGK  FLAGG1  FLAGG2  FLAGG3  GKENRMNT GKCHAPT1
##     <dbl> <dbl+lbl>   <dbl+lb> <dbl+l> <dbl+l> <dbl+l> <dbl+l>    <dbl> <dbl+lb>
##  1 112038 4 [URBAN]   5 [K-5]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      330 1 [YES]
##  2 123056 3 [RURAL]   8 [K-8]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      620 1 [YES]
##  3 128068 3 [RURAL]   8 [K-8]  1 [Yes] 0 [No]  0 [No]  0 [No]       540 1 [YES]
##  4 128076 2 [SUBURBA… 8 [K-8]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      564 2 [NO]
##  5 128079 3 [RURAL]   5 [K-5]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      471 1 [YES]
##  6 130085 2 [SUBURBA… 5 [K-5]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      521 1 [YES]
##  7 159171 3 [RURAL]   5 [K-5]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      645 1 [YES]
##  8 161176 3 [RURAL]   3 [K-3]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      444 1 [YES]
##  9 161183 3 [RURAL]   3 [K-3]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      662 1 [YES]
## 10 162184 3 [RURAL]   8 [K-8]  1 [Yes] 1 [Yes] 1 [Yes] 1 [Yes]      650 1 [YES]
## # ℹ 70 more rows
## # ℹ 44 more variables: GKFRLNCH <dbl>, GKBUSED <dbl>, GKNATVAM <dbl>,
## #   GKASIAN <dbl>, GKBLACK <dbl>, GKHSPANC <dbl>, GKWHITE <dbl>,
## #   GKOTHRAC <dbl>, G1ENRMNT <dbl>, G1AVGDAT <dbl>, G1AVGDMB <dbl>,
## #   G1CHAPT1 <dbl+lbl>, G1FRLNCH <dbl>, G1BUSED <dbl>, G1NATVAM <dbl>,
## #   G1ASIAN <dbl>, G1BLACK <dbl>, G1HSPANC <dbl>, G1WHITE <dbl>,
## #   G1OTHRAC <dbl>, G2ENRMNT <dbl>, G2AVGDAT <dbl>, G2AVGDMB <dbl>, …
```

```
students <- read_sav("STAR_Students.sav")
```

```
## Failed to find G1READ_A
## Failed to find G1MATH_A
## Failed to find G2READ_A
## Failed to find G2MATH_A
## Failed to find G3READ_A
## Failed to find G3MATH_A
```

```
students
```

```
## # A tibble: 11,601 × 379
##    stdntid gender  race    birthmonth birthday birthyear FLAGSGK FLAGSG1 FLAGSG2
##      <dbl> <dbl+l> <dbl+l> <dbl+lbl>     <dbl>     <dbl> <dbl+l> <dbl+l> <dbl+l>
##  1   10000 1 [MAL… 1 [WHI… 1 [JANUAR…       22      1979 0 [NO]  1 [YES] 1 [YES]
##  2   10001 1 [MAL… 1 [WHI… 2 [FEBRUA…       20      1980 1 [YES] 0 [NO]  0 [NO]
##  3   10002 2 [FEM… 2 [BLA… 7 [JULY]         21      1979 0 [NO]  0 [NO]  0 [NO]
##  4   10003 1 [MAL… 1 [WHI… 5 [MAY]          28      1980 0 [NO]  1 [YES] 1 [YES]
##  5   10004 2 [FEM… 2 [BLA… 1 [JANUAR…        2      1980 0 [NO]  0 [NO]  1 [YES]
##  6   10005 1 [MAL… 2 [BLA… 1 [JANUAR…        1      1980 0 [NO]  0 [NO]  1 [YES]
##  7   10006 2 [FEM… 2 [BLA… 2 [FEBRUA…        5      1979 0 [NO]  0 [NO]  1 [YES]
##  8   10007 1 [MAL… 2 [BLA… 3 [MARCH]        15      1980 0 [NO]  0 [NO]  0 [NO]
##  9   10008 2 [FEM… 2 [BLA… 4 [ARPIL]        23      1979 0 [NO]  1 [YES] 1 [YES]
## 10   10009 2 [FEM… 2 [BLA… 4 [ARPIL]        24      1980 0 [NO]  0 [NO]  0 [NO]
## # ℹ 11,591 more rows
## # ℹ 370 more variables: FLAGSG3 <dbl+lbl>, flaggk <dbl+lbl>, flagg1 <dbl+lbl>,
## #   flagg2 <dbl+lbl>, flagg3 <dbl+lbl>, flagg4 <dbl+lbl>, flagg5 <dbl+lbl>,
## #   flagg6 <dbl+lbl>, flagg7 <dbl+lbl>, flagg8 <dbl+lbl>, flagprt4 <dbl+lbl>,
## #   flagidn8 <dbl+lbl>, flagprt8 <dbl+lbl>, flagsatact <dbl+lbl>,
## #   flaghscourse <dbl+lbl>, flaghsgraduate <dbl+lbl>, gkclasstype <dbl+lbl>,
## #   g1classtype <dbl+lbl>, g2classtype <dbl+lbl>, g3classtype <dbl+lbl>, …
```

Bin socioeconomic status proportions

```
summary(k3_schools$SCHLURBN)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   2.000   3.000   2.462   3.000   4.000
```

```
table(k3_schools$SCHLURBN)
```

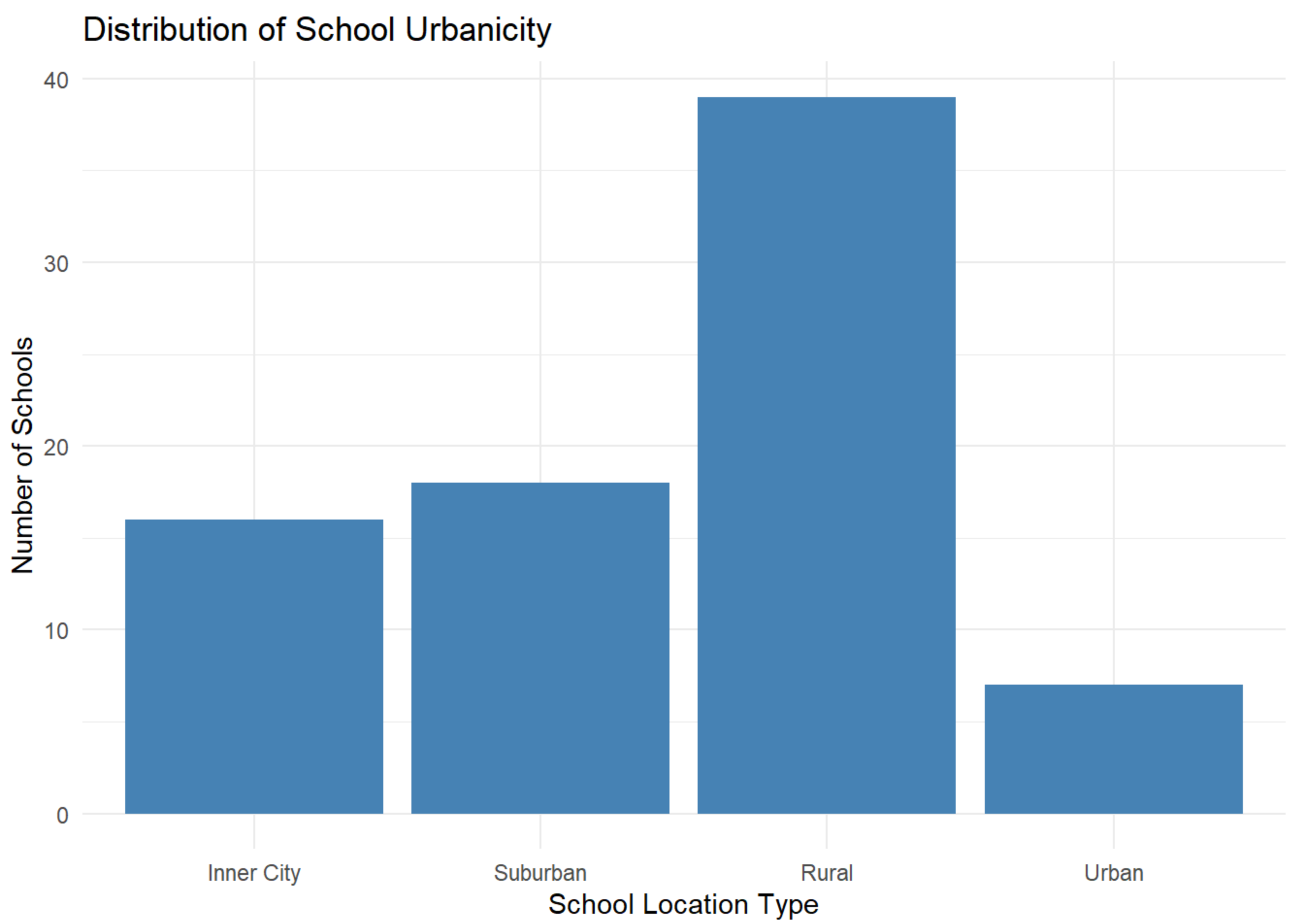
```
## 
##  1  2  3  4 
## 16 18 39  7
```

```
k3_schools <- k3_schools %>%
  mutate(
    school_urban = factor(
      SCHLURBN,
      levels = c(1,2,3,4),
      labels = c("Inner City", "Suburban", "Rural", "Urban")
    )
  )
```

```
# Frequency table
table(k3_schools$school_urban)
```

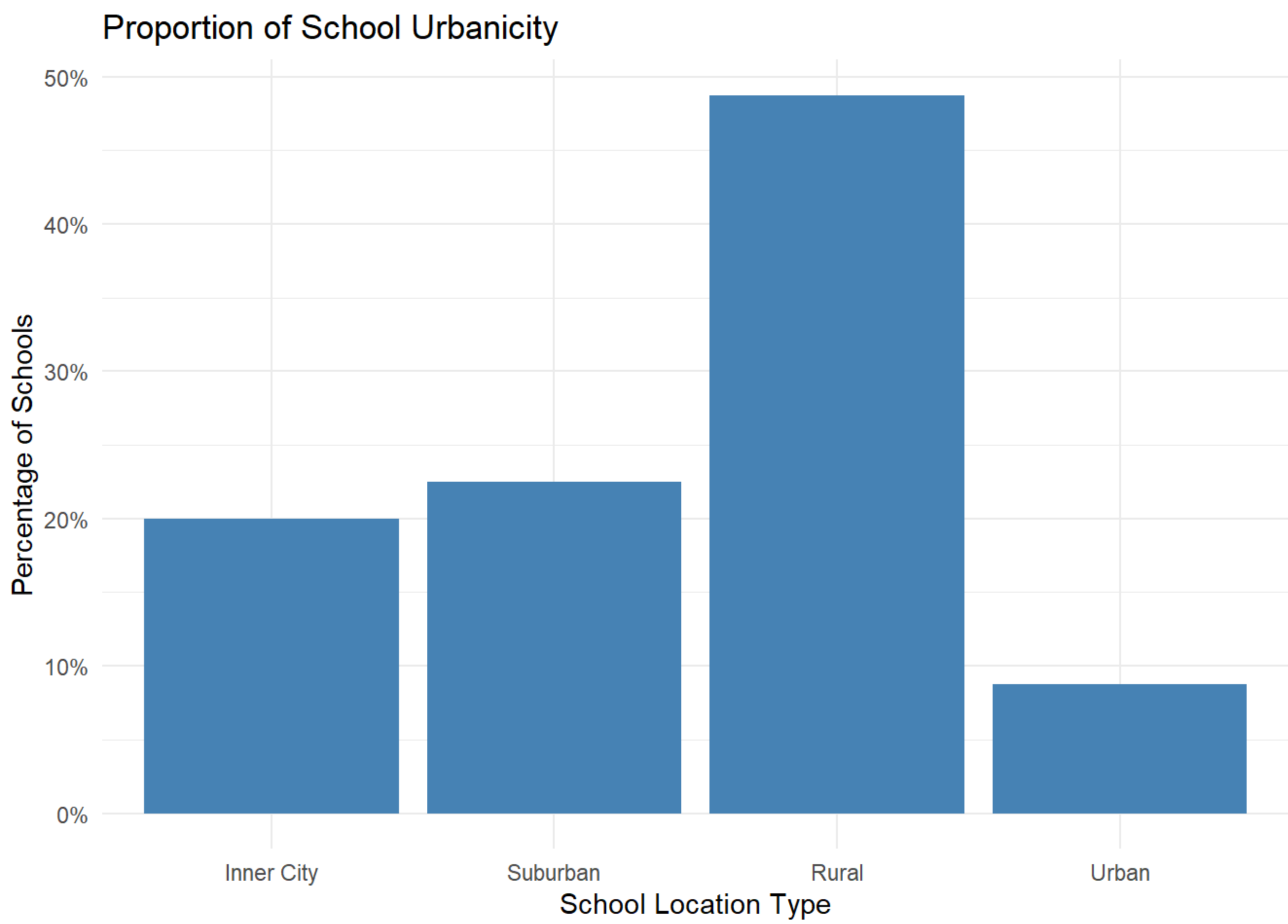
```
##
## Inner City   Suburban      Rural      Urban
##         16         18         39          7
```

```
#dist of school urbanicity
ggplot(k3_schools, aes(x = school_urban)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Distribution of School Urbanicity",
    x = "School Location Type",
    y = "Number of Schools"
  ) +
  theme_minimal()
```



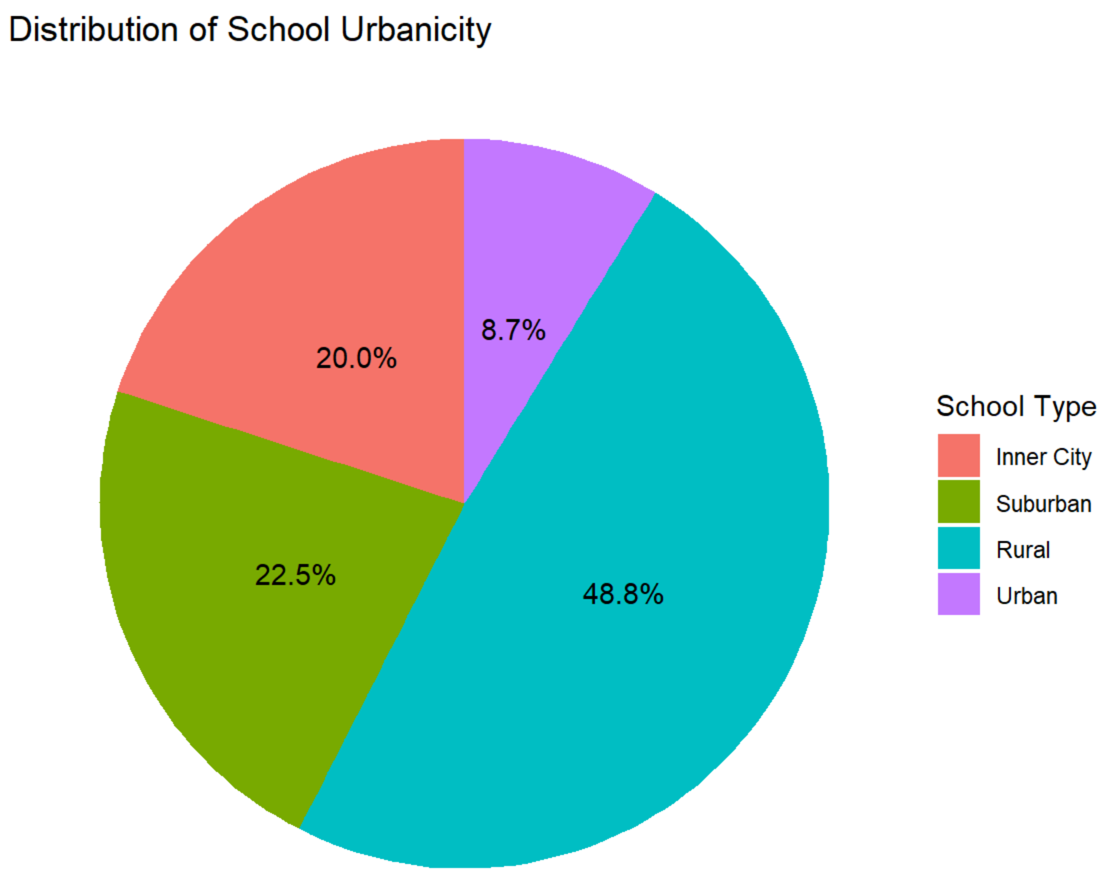
```
# percent table
ggplot(k3_schools, aes(x = school_urban)) +
  geom_bar(aes(y = (..count..)/sum(..count..)), fill = "steelblue") +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Proportion of School Urbanicity",
    x = "School Location Type",
    y = "Percentage of Schools"
  ) +
  theme_minimal()
```

```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## ℹ Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
urban_dist <- k3_schools %>%
  count(school_urban) %>%
  mutate(percent = n / sum(n))
```

```
#pie chart
ggplot(urban_dist, aes(x = "", y = percent, fill = school_urban)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  geom_text(aes(label = scales::percent(percent)),
            position = position_stack(vjust = 0.5)) +
  labs(
    title = "Distribution of School Urbanicity",
    fill = "School Type"
  ) +
  theme_void()
```



lunch

```
summary(k3_schools$GKFRLNCH)
```

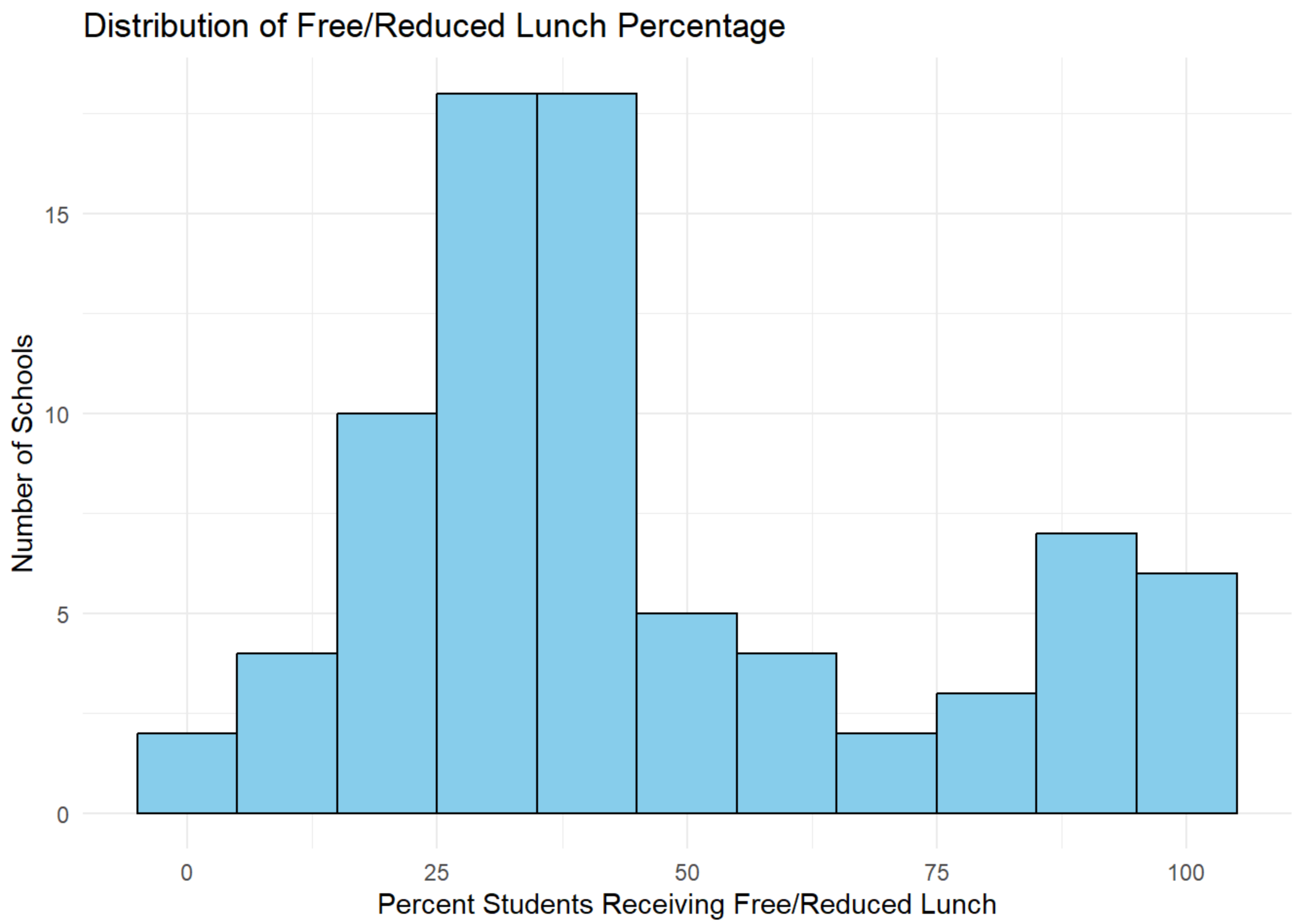
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##    1.00   29.00   38.00   46.37   58.50   99.00       1
```

```
sum(is.na(k3_schools$GKFRLNCH))
```

```
## [1] 1
```

```
ggplot(k3_schools, aes(x = GKFRLNCH)) +
  geom_histogram(binwidth = 10, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Free/Reduced Lunch Percentage",
    x = "Percent Students Receiving Free/Reduced Lunch",
    y = "Number of Schools"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_bin()`).
```



Bin socioeconomic status proportions

```
#Higher percent, Low Socioeconomic Status
k3_schools <- k3_schools %>%
  mutate(
    ses_bin = cut(
      GKFRLNCH,
      breaks = c(0,25,50,75,100),
      labels = c("High SES","Middle SES","Low SES","Very Low SES"),
      include.lowest = TRUE
    )
  )
```

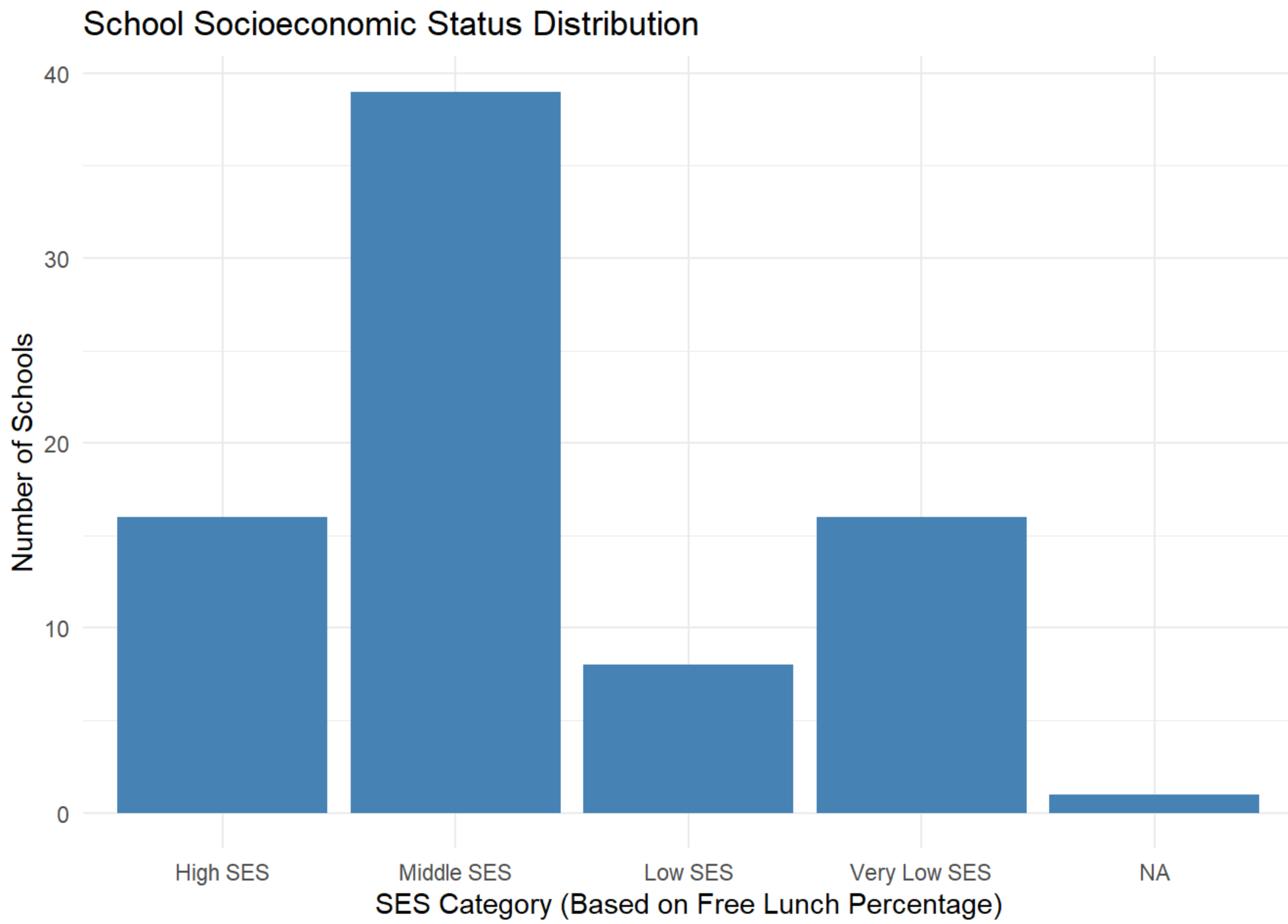
```
table(k3_schools$ses_bin)
```

```
##
##     High SES   Middle SES      Low SES Very Low SES
##           16           39            8           16
```

```
k3_schools %>%
  count(ses_bin) %>%
  mutate(percent = n/sum(n)*100)
```

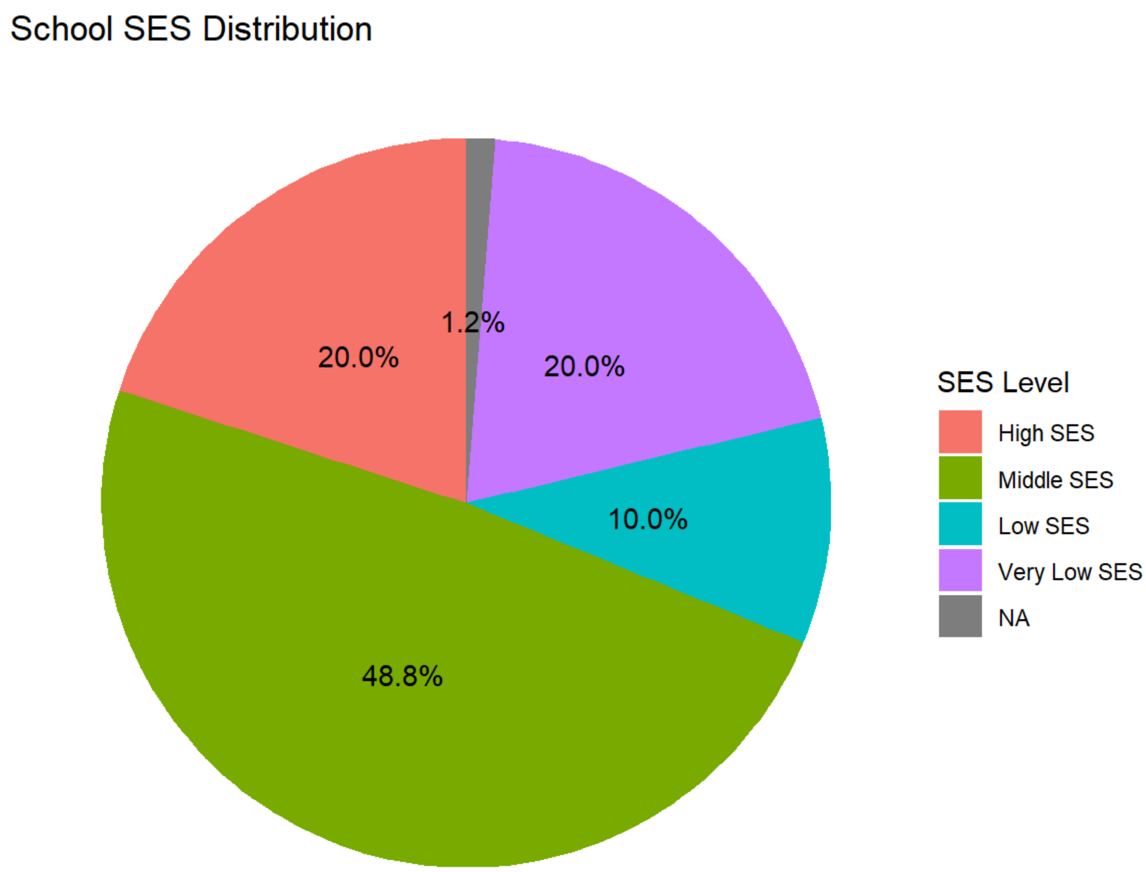
```
## # A tibble: 5 × 3
##   ses_bin          n percent
##   <fct>        <int>   <dbl>
## 1 High SES        16   20
## 2 Middle SES      39   48.8
## 3 Low SES          8   10
## 4 Very Low SES    16   20
## 5 <NA>             1    1.25
```

```
#bar chart
ggplot(k3_schools, aes(x = ses_bin)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "School Socioeconomic Status Distribution",
    x = "SES Category (Based on Free Lunch Percentage)",
    y = "Number of Schools"
  ) +
  theme_minimal()
```




```
# pie chart
ses_dist <- k3_schools %>%
  count(ses_bin) %>%
  mutate(percent = n/sum(n))

ggplot(ses_dist, aes(x="", y=percent, fill=ses_bin)) +
  geom_col(width=1) +
  coord_polar(theta="y") +
  geom_text(aes(label=scales::percent(percent)),
            position=position_stack(vjust=0.5)) +
  labs(
    title="School SES Distribution",
    fill="SES Level"
  ) +
  theme_void()
```



Grades (gtmathss, gtreadss) vs gender

grade k math

```
summary(students$gktmathss)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   288.0   454.0   484.0   485.4   513.0   626.0    5730
```

```
summary(students$gktreadss)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   315.0   414.0   433.0   436.7   453.0   627.0    5812
```

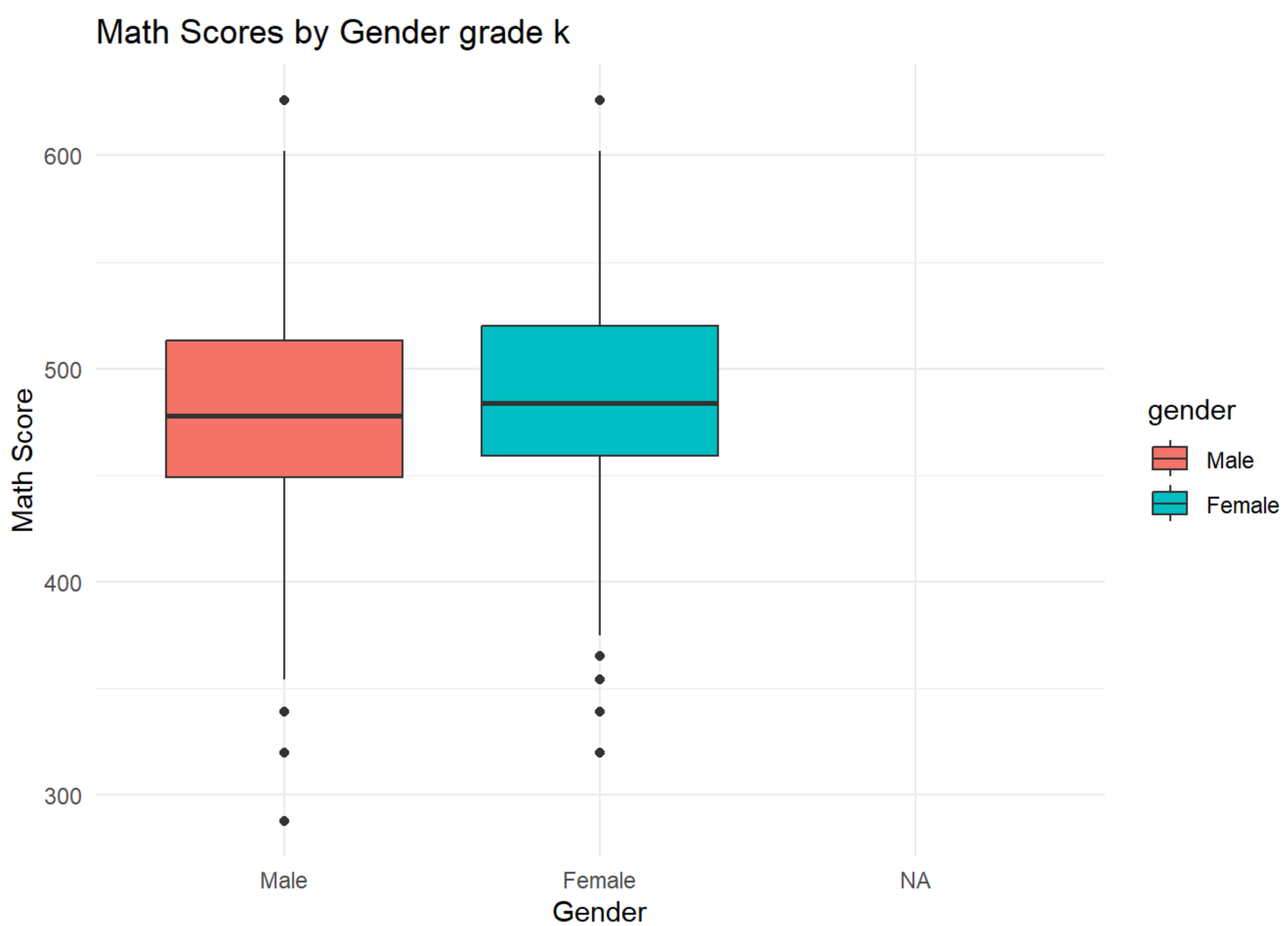
```
table(students$gender)
```

```
## 
##    1    2 
## 6124 5457
```

```
students$gender <- factor(
  students$gender,
  levels = c(1,2),
  labels = c("Male","Female")
)
```

```
ggplot(students, aes(x = gender, y = gktmathss, fill = gender)) +
  geom_boxplot() +
  labs(
    title = "Math Scores by Gender grade k",
    x = "Gender",
    y = "Math Score"
  ) +
  theme_minimal()
```

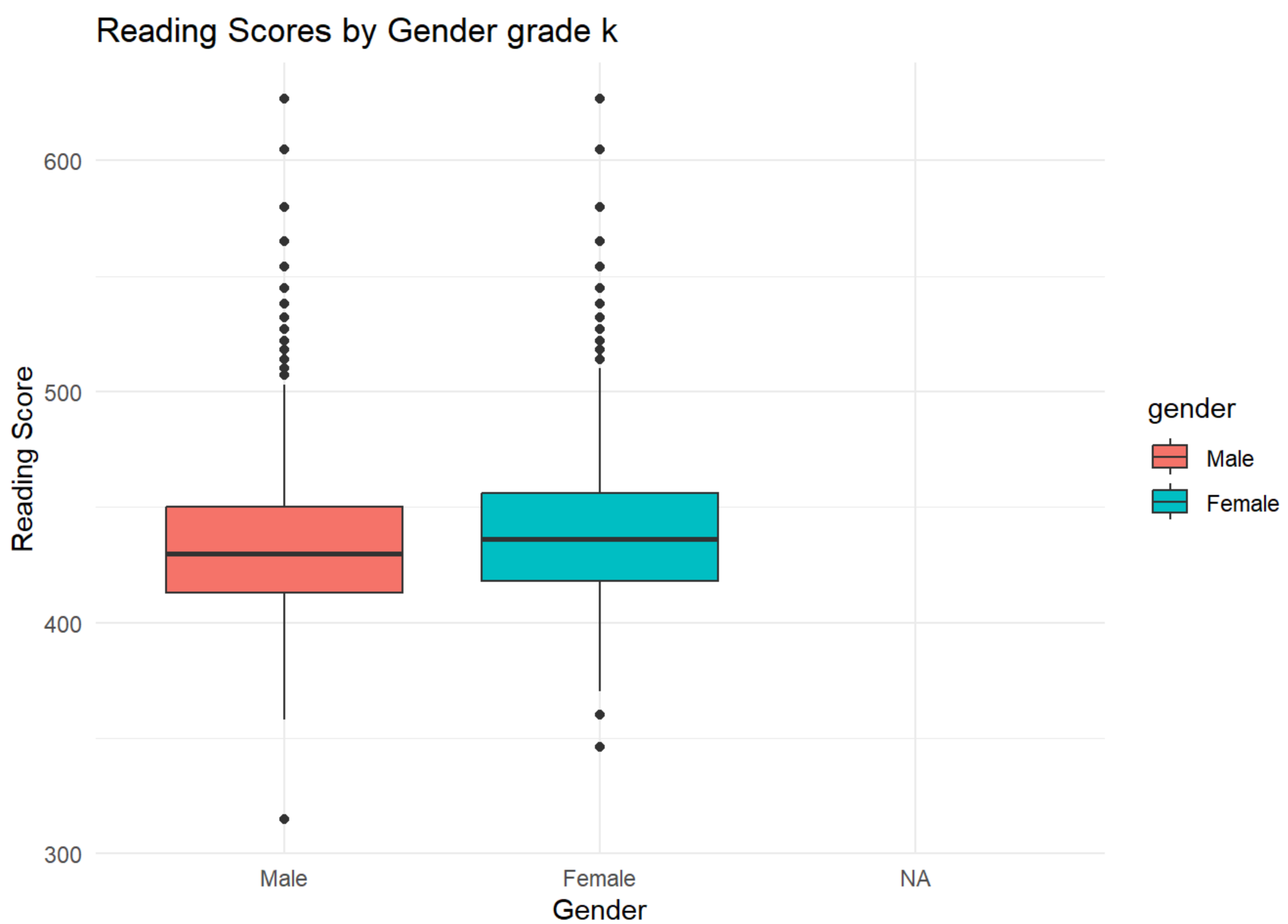
```
## Warning: Removed 5730 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



grade k reading

```
ggplot(students, aes(x = gender, y = gktreadss, fill = gender)) +
  geom_boxplot() +
  labs(
    title = "Reading Scores by Gender grade k",
    x = "Gender",
    y = "Reading Score"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5812 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
students %>%
  group_by(gender) %>%
  summarise(
    mean_math = mean(gktmathss, na.rm = TRUE),
    mean_read = mean(gktreadss, na.rm = TRUE)
  )
```

```
## # A tibble: 3 × 3
##   gender mean_math mean_read
##   <fct>      <dbl>     <dbl>
## 1 Male        482.      434.
## 2 Female      489.      440.
## 3 <NA>        NaN       NaN
```

k-3 math in one chart

```
library(dplyr)
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

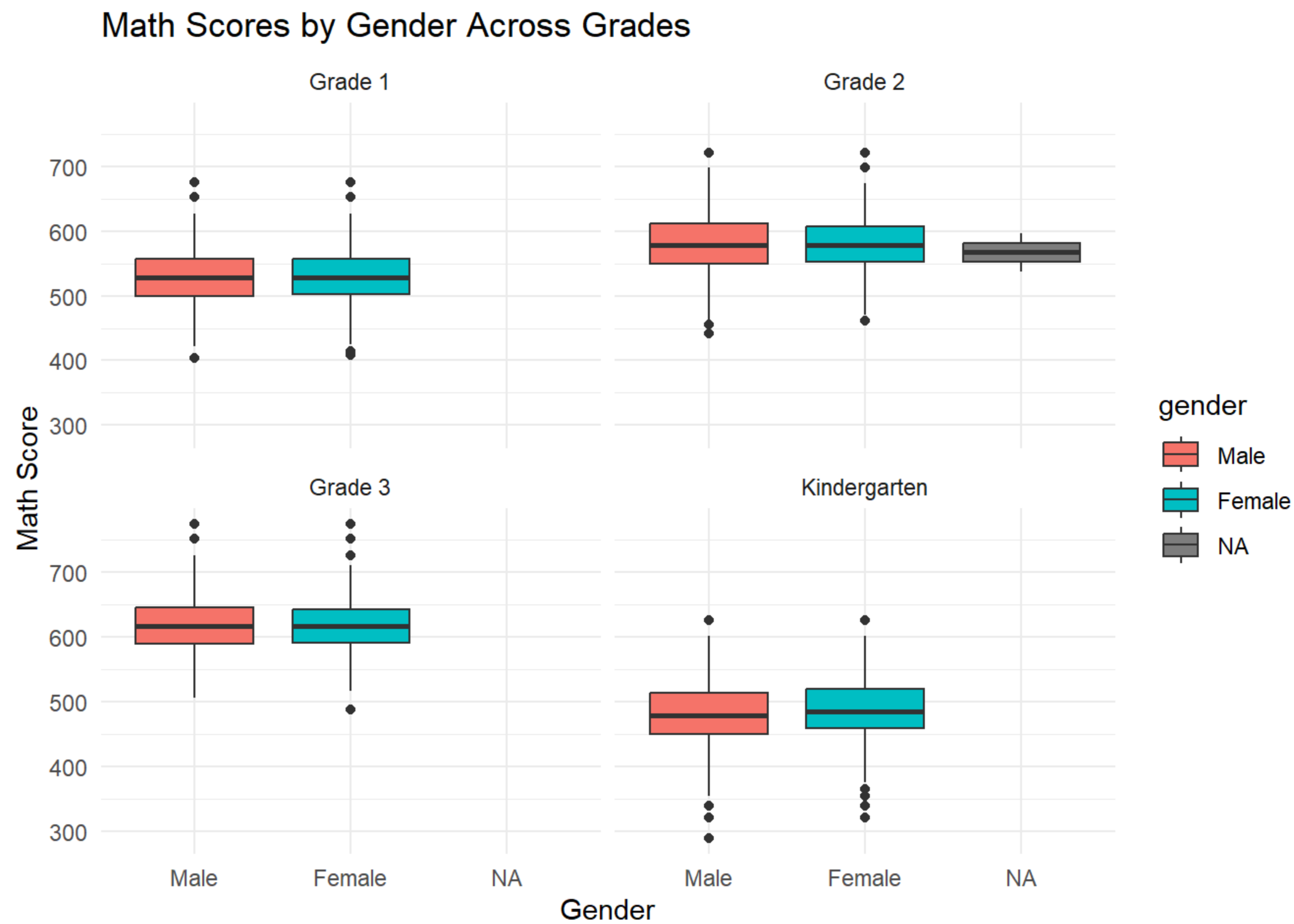
```
library(ggplot2)
long_data <- students %>%
  pivot_longer(
    cols = c(gktmathss, g1tmathss, g2tmathss, g3tmathss),
    names_to = "grade",
    values_to = "math_score"
  ) %>%
  mutate(
    gender = factor(gender),
    grade = recode(grade,
      "gktmathss" = "Kindergarten",
      "g1tmathss" = "Grade 1",
      "g2tmathss" = "Grade 2",
      "g3tmathss" = "Grade 3"
    )
  )
```

```
table(long_data$grade)
```

```
##
##      Grade 1      Grade 2      Grade 3 Kindergarten
##        11601        11601        11601        11601
```

```
ggplot(long_data, aes(x = gender, y = math_score, fill = gender)) +
  geom_boxplot() +
  facet_wrap(~grade) +
  labs(
    title = "Math Scores by Gender Across Grades",
    x = "Gender",
    y = "Math Score"
  ) +
  theme_minimal()
```

```
## Warning: Removed 21793 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



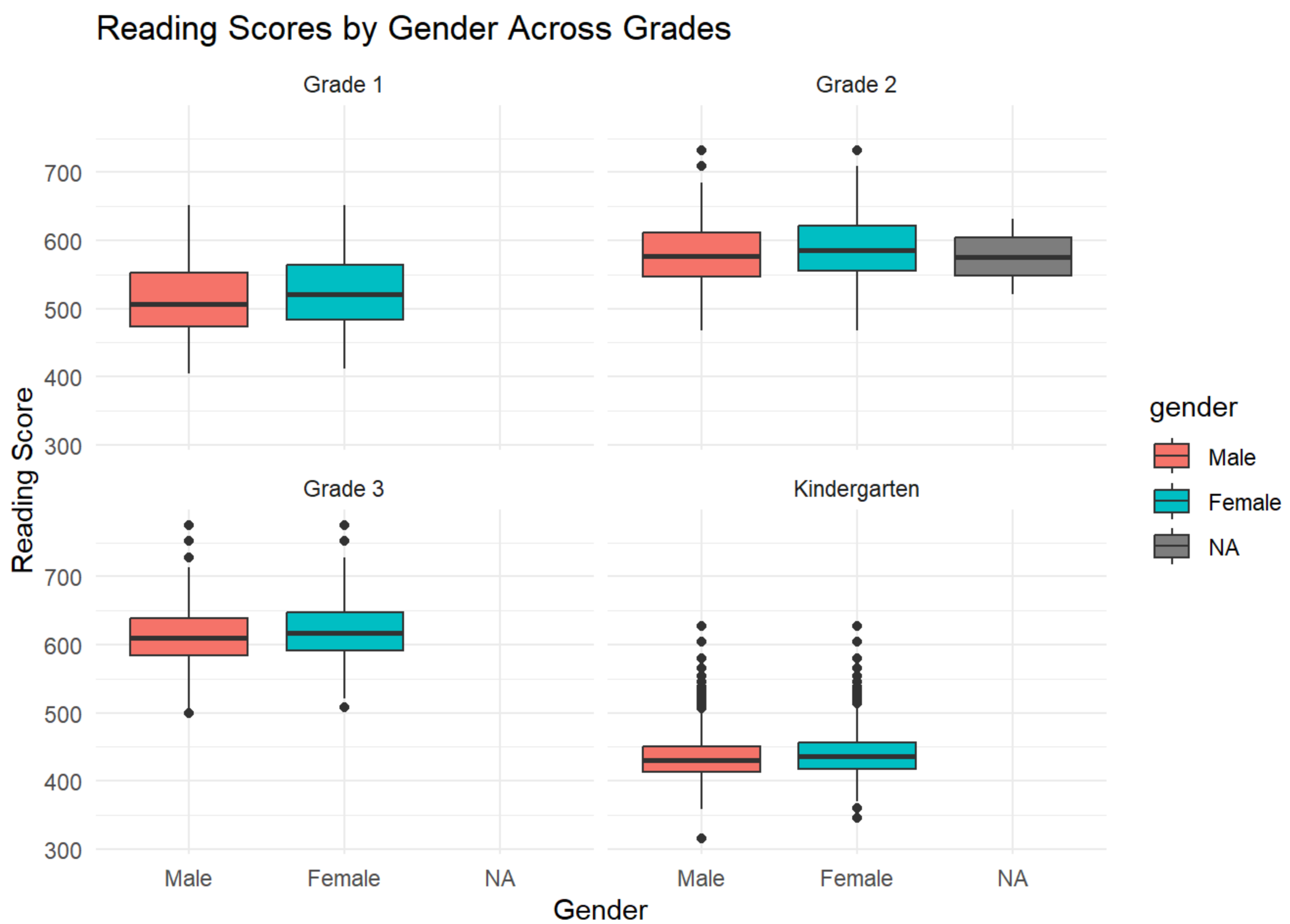
k-3 reading

```
### K-3 Reading in one chart
long_read <- students %>%
  mutate(gender = factor(gender)) %>%
  pivot_longer(
    cols = c(gktreadss, g1treadss, g2treadss, g3treadss),
    names_to = "grade",
    values_to = "read_score"
  )
```

```
long_read <- long_read %>%
  mutate(grade = recode(grade,
    gktreadss = "Kindergarten",
    g1treadss = "Grade 1",
    g2treadss = "Grade 2",
    g3treadss = "Grade 3"
  ))
```

```
ggplot(long_read, aes(x = gender, y = read_score, fill = gender)) +
  geom_boxplot() +
  facet_wrap(~grade) +
  labs(
    title = "Reading Scores by Gender Across Grades",
    x = "Gender",
    y = "Reading Score"
  ) +
  theme_minimal()
```

```
## Warning: Removed 22143 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



grade math vs race

```
table(students$race)
```

```
##
##    1    2    3    4    5    6
## 7200 4180   32   21   14   20
```

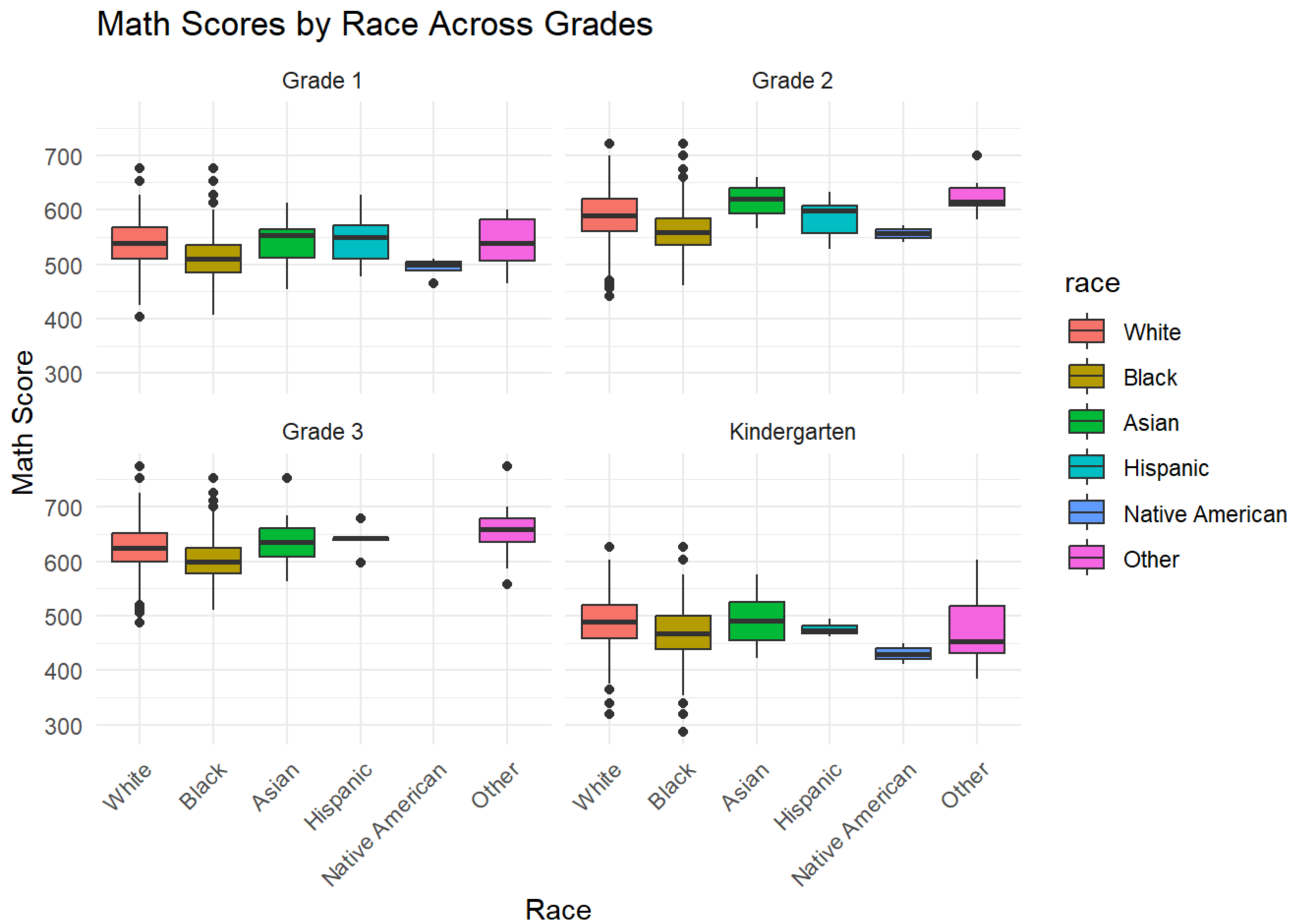
```
students$race <- factor(
  students$race,
  levels = c(1,2,3,4,5,6),
  labels = c("White","Black","Asian","Hispanic","Native American","Other")
)
```

```
long_math_race <- students %>%
  pivot_longer(
    cols = c(gktmathss, g1tmathss, g2tmathss, g3tmathss),
    names_to = "grade",
    values_to = "math_score"
  ) %>%
  filter(!is.na(race))
```

```
long_math_race <- long_math_race %>%
  mutate(grade = recode(grade,
    gktmathss = "Kindergarten",
    g1tmathss = "Grade 1",
    g2tmathss = "Grade 2",
    g3tmathss = "Grade 3"
  ))
```

```
ggplot(long_math_race, aes(x = race, y = math_score, fill = race)) +
  geom_boxplot() +
  facet_wrap(~grade) +
  labs(
    title = "Math Scores by Race Across Grades",
    x = "Race",
    y = "Math Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

```
## Warning: Removed 21285 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```


Math Scores by Race Across Grades
Grade 1
Grade 2
Grade 3
Kindergarten
Math Score
700
600
500
400
300
White
Black
Asian
Hispanic
Native American
Other
Race
race
White
Black
Asian
Hispanic
Native American
Other

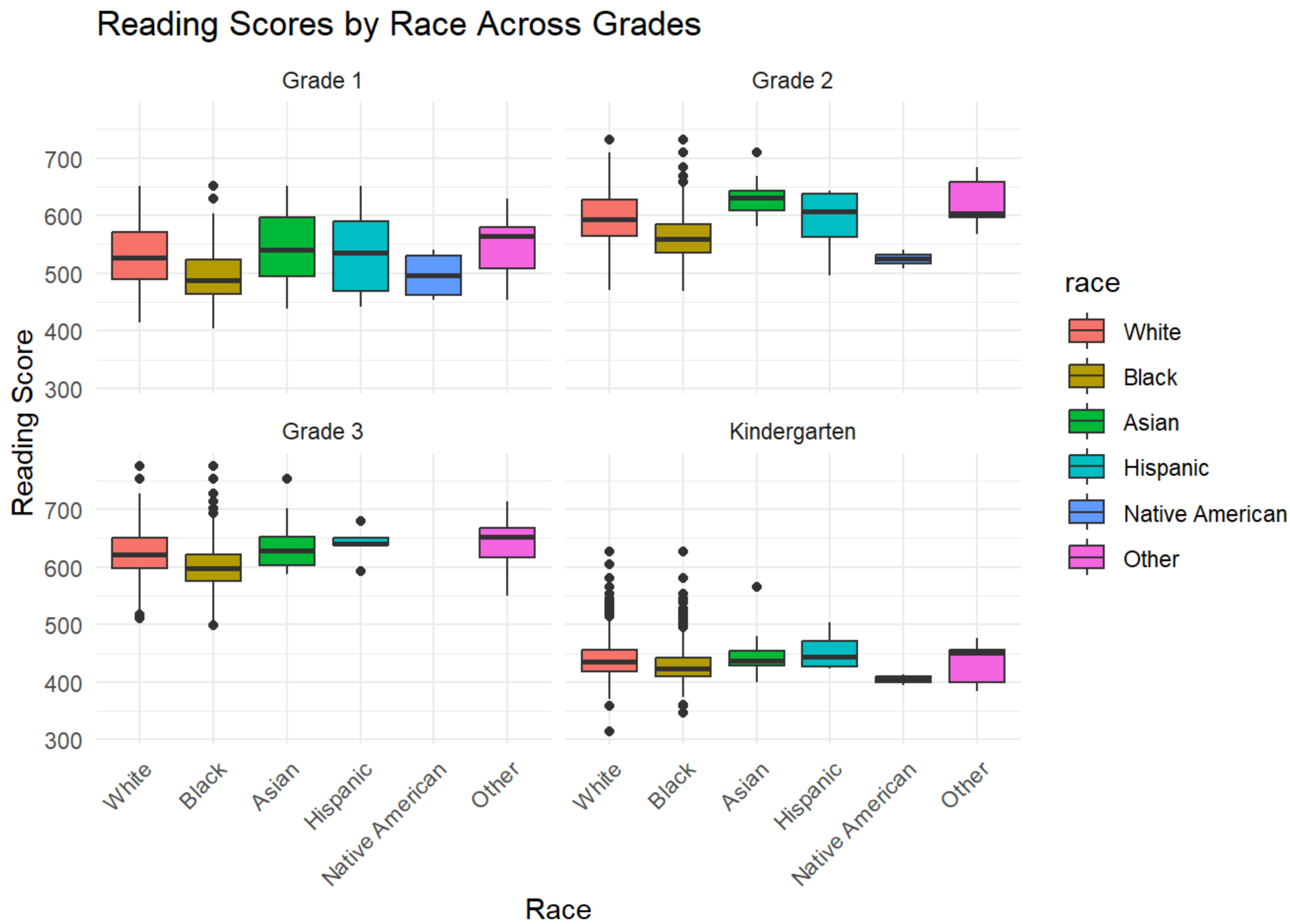
reading vs race

```
long_read_race <- students %>%
  pivot_longer(
    cols = c(gktreadss, g1treadss, g2treadss, g3treadss),
    names_to = "grade",
    values_to = "read_score"
  ) %>%
  filter(!is.na(race))

long_read_race <- long_read_race %>%
  mutate(
    grade = recode(grade,
      gktreadss = "Kindergarten",
      g1treadss = "Grade 1",
      g2treadss = "Grade 2",
      g3treadss = "Grade 3"
    )
  )

ggplot(long_read_race, aes(x = race, y = read_score, fill = race)) +
  geom_boxplot() +
  facet_wrap(~grade) +
  labs(
    title = "Reading Scores by Race Across Grades",
    x = "Race",
    y = "Reading Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

```
## Warning: Removed 21636 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Average grade per class size groups

```
students <- students %>%
  mutate(
    gkclasst = factor(gkclasstype, levels=c(1,2,3),
                      labels=c("Small","Regular","Regular+Aide")),
    g1classt = factor(g1classtype, levels=c(1,2,3),
                      labels=c("Small","Regular","Regular+Aide")),
    g2classt = factor(g2classtype, levels=c(1,2,3),
                      labels=c("Small","Regular","Regular+Aide")),
    g3classt = factor(g3classtype, levels=c(1,2,3),
                      labels=c("Small","Regular","Regular+Aide"))
  )
```

```
students <- students %>%
  mutate(
    small_years =
      (gkclasst=="Small") +
      (g1classt=="Small") +
      (g2classt=="Small") +
      (g3classt=="Small")
  )
```

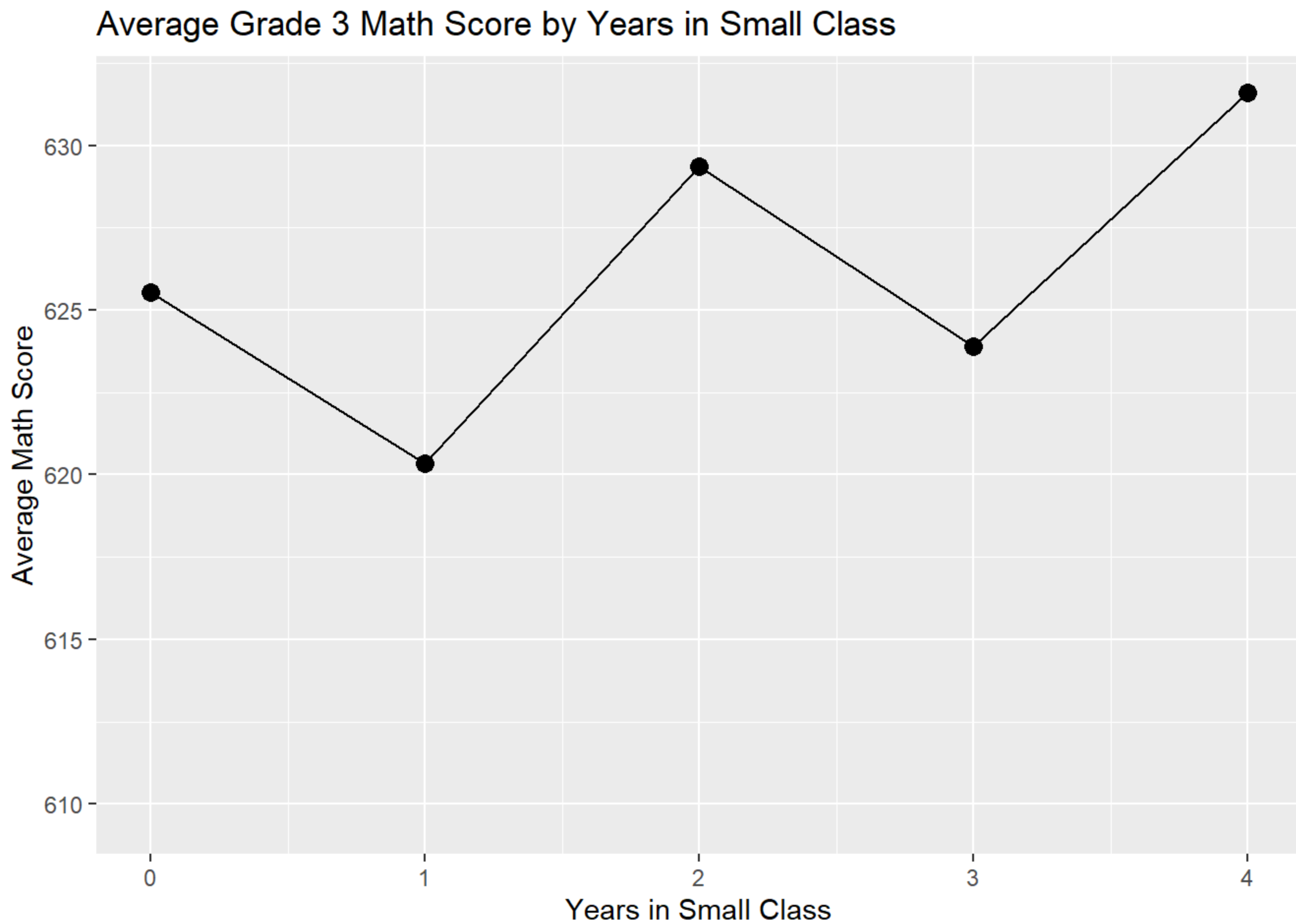
```
# since grade 3 is the end of our test, so i will use grade3 grade as response.
students %>%
  group_by(small_years) %>%
  summarise(
    avg_math = mean(g3tmathss, na.rm=TRUE),
    avg_read = mean(g3treadss, na.rm=TRUE)
  )
```

```
## # A tibble: 6 × 3
##   small_years avg_math avg_read
##         <int>    <dbl>    <dbl>
## 1           0     626.     623.
## 2           1     620.     618.
## 3           2     629.     624.
## 4           3     624.     620.
## 5           4     632.     631.
## 6          NA     610.     607.
```

```
students %>%
  group_by(small_years) %>%
  summarise(avg_math = mean(g3tmathss, na.rm=TRUE)) %>%
  ggplot(aes(x = small_years, y = avg_math)) +
  geom_point(size=3) +
  geom_line() +
  labs(
    title="Average Grade 3 Math Score by Years in Small Class",
    x="Years in Small Class",
    y="Average Math Score"
  )
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```




```
students %>%
  group_by(small_years) %>%
  summarise(avg_read = mean(g3treadss, na.rm = TRUE)) %>%
  ggplot(aes(x = small_years, y = avg_read)) +
  geom_point(size = 3) +
  geom_line() +
  labs(
    title = "Average Grade 3 Reading Score by Years in Small Class",
    x = "Years in Small Class",
    y = "Average Reading Score"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

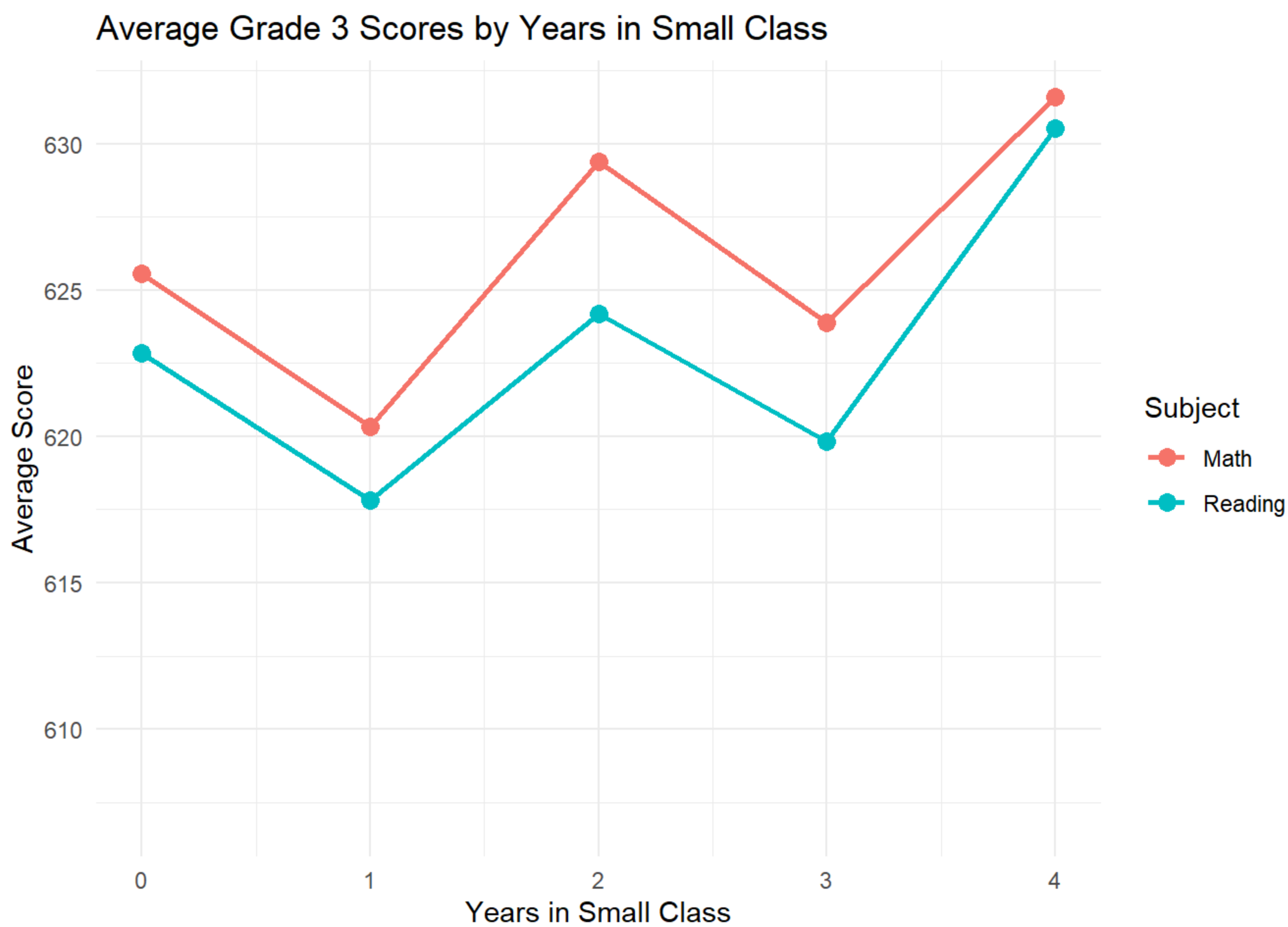

Average Grade 3 Reading Score by Years in Small Class
Average Reading Score
Years in Small Class
630
625
620
615
610
0
1
2
3
4

```r
library(dplyr)
library(tidyr)
library(ggplot2)

students %>%
  group_by(small_years) %>%
  summarise(
    avg_math = mean(g3tmathss, na.rm = TRUE),
    avg_read = mean(g3treadss, na.rm = TRUE)
  ) %>%
  pivot_longer(
    cols = c(avg_math, avg_read),
    names_to = "subject",
    values_to = "avg_score"
  ) %>%
  mutate(subject = recode(subject,
                          avg_math = "Math",
                          avg_read = "Reading")) %>%
  ggplot(aes(x = small_years, y = avg_score, color = subject)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(
    title = "Average Grade 3 Scores by Years in Small Class",
    x = "Years in Small Class",
    y = "Average Score",
    color = "Subject"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



Average Grade 3 Scores by Years in Small Class
Average Score
630
625
620
615
610
0
1
2
3
4
Years in Small Class
Subject
Math
Reading